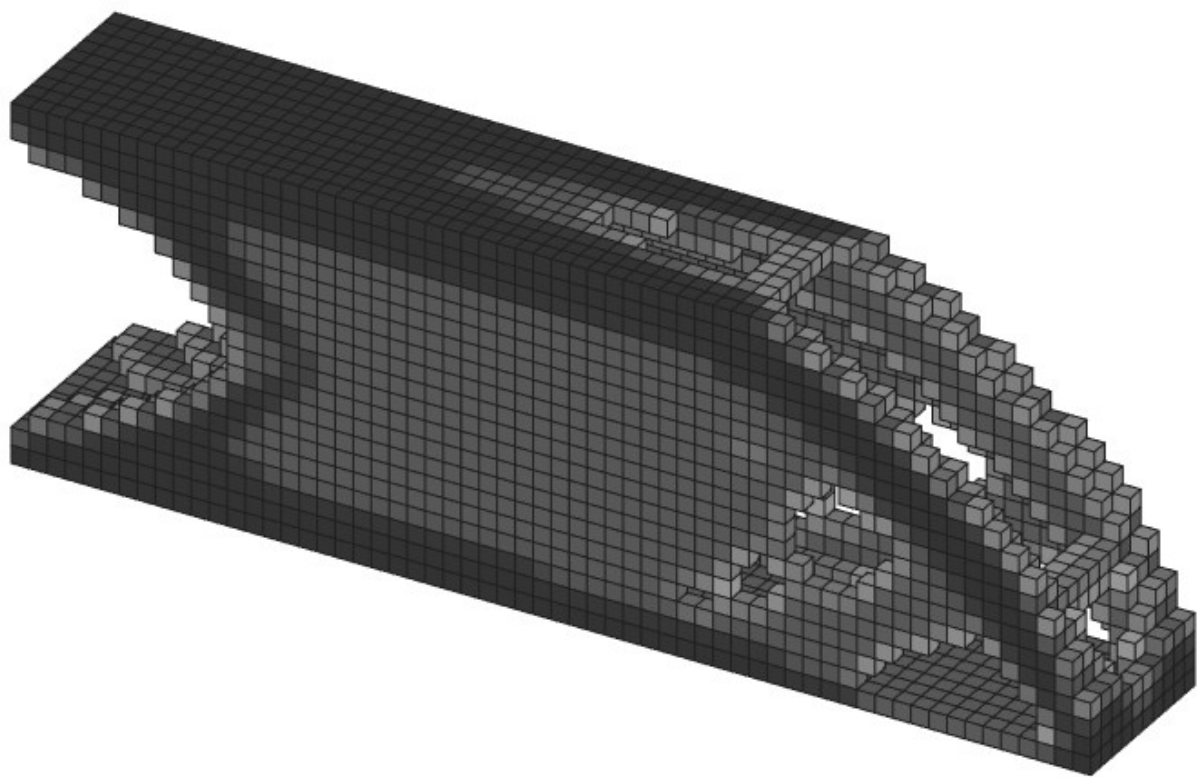


Topology Optimization for 3D Structures Design

A simulation-led design study exploring how geometry and loading conditions shape lightweight, high-stiffness 3D structural forms.



Role Researcher, computational designer, MATLAB simulation, analysis, visualisation	Methods SIMP topology optimization, finite element analysis, density filtering	Tools MATLAB, modified top3d.m workflow, 3D voxel visualisation
Scope 3 geometry ratios + 4 loading profiles	Goal Reduce material while maintaining stiffness and clear load paths	Output Optimized 3D structural topologies for portfolio and engineering review

Project summary

This project converts an engineering dissertation into a portfolio-ready computational design case study. It investigates how 3D structural forms can be made lighter and stiffer by optimizing where material is placed inside a design domain. The study uses topology optimization to remove non-critical material while preserving efficient load paths, producing organic, voxel-based structural forms that are well suited to advanced manufacturing and concept-stage mechanical design.

The work focuses on two design questions: how aspect ratio changes influence the optimized geometry, and how different loading profiles alter the final structural topology. The result is a clear visual comparison of geometry-driven and load-driven design behaviour across multiple 3D cases.

Design challenge

Many topology optimization examples are demonstrated in 2D because they are easier to visualize and faster to compute. Real engineering components, however, behave in 3D, where load paths, stiffness, void formation, and manufacturability become more complex. The challenge was to create a controlled 3D workflow that makes these relationships visible, comparable, and useful for design decision-making.

Objectives

- Use an accessible MATLAB-based topology optimization workflow to explore 3D structural design.
- Compare how three aspect ratios affect material distribution, compliance, and stiffness behaviour.
- Compare how central, uniform, triangular, and edge loading profiles shape structural load paths.
- Translate simulation outputs into visually readable design evidence for a professional portfolio.

Methodology

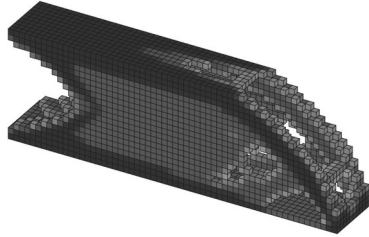
The project used the SIMP method with finite element analysis to iteratively update element densities. A fixed support was applied to one side of the design domain while load cases were applied to the opposite side or selected surfaces. Each simulation produced a final density-based topology, allowing visual and qualitative comparison of material efficiency, compliance behaviour, and robustness.

1. Define domain Set mesh dimensions and boundary conditions for each design case.	2. Apply load Test geometry ratios and load distributions under controlled conditions.	3. Optimize density Use SIMP iterations to remove inefficient material and preserve useful load paths.	4. Compare outcomes Assess compliance, topology shape, material efficiency, and robustness.
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CASE A - GEOMETRY VARIATION

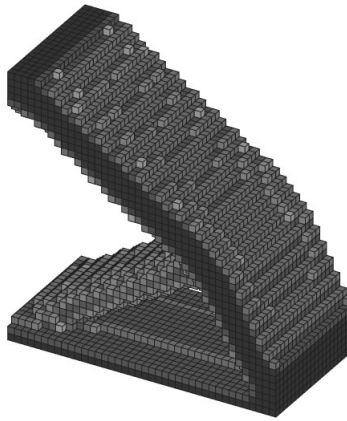
How aspect ratio changes the optimized form

Case A studies three design domains with different length, breadth, and width ratios. The comparison shows that geometry is not only a container for the structure; it actively shapes the direction of force flow, the amount of bending, and the type of bracing produced by the optimizer.



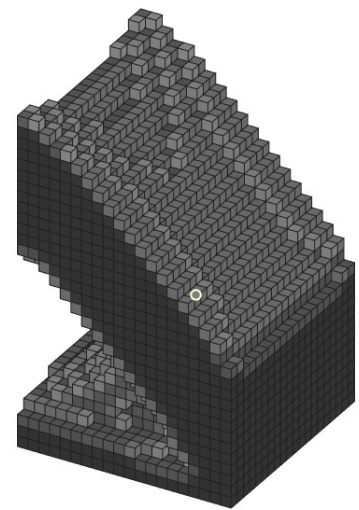
A1 - Elongated domain / 3:1:0.5

Produced long axial members and diagonal bracing. This was the most efficient geometry in the study, with material aligned along the main load path.



A2 - Balanced domain / 2:2:1

Developed a more distributed lattice-like structure. It offers balanced performance with multiple load paths and moderate compliance.



A3 - Tall/slim domain / 1:3:1

Formed vertical reinforcement and plate-like bracing. The taller profile increased bending sensitivity and reduced stiffness efficiency.

Design insight

The elongated geometry created the cleanest stiffness-to-weight response because its dimensions aligned with the principal load path. The balanced geometry created more redundancy, while the tall/slim geometry required extra bracing to resist bending. This shows why early-stage proportion decisions can strongly influence optimized structural quality.

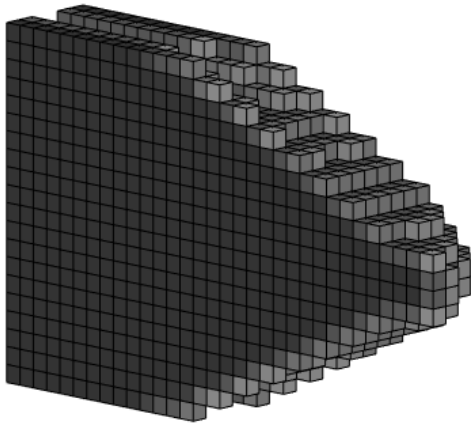
Portfolio caption

A comparative set of 3D topology-optimized structures showing how aspect ratio influences structural load paths, void formation, stiffness, and material efficiency.

CASE B - LOADING VARIATION

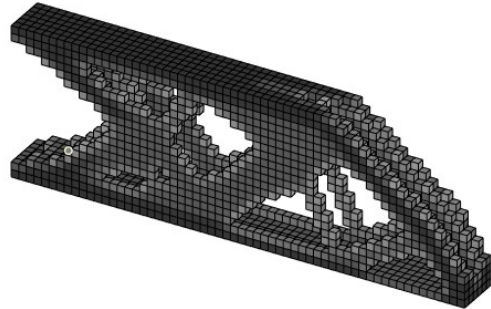
How force distribution shapes topology

Case B keeps the geometry constant and changes the way external force is applied. Each load profile creates a different structural logic: point loads generate localized reinforcement, distributed loads create broad webs, triangular loads create asymmetric density gradients, and edge loads generate diagonal buttress-like support paths.



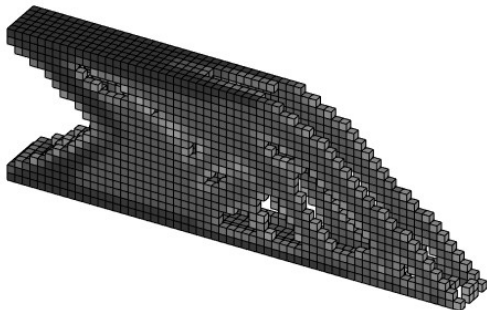
B1 - Central point load

A localized point force produced a concentrated, stepped load path. Efficient for the defined load but less robust under off-axis changes.



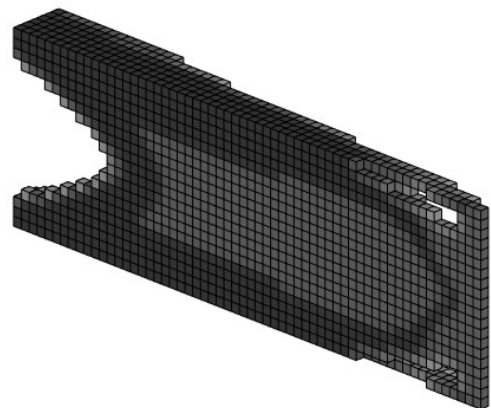
B2 - Uniform distributed load

A broad pressure load produced fan-shaped webs and dispersed voids, spreading stiffness across the structure.



B3 - Triangular distributed load

A linearly varying load created an asymmetric material layout, with more density toward the high-pressure side.



B4 - Edge-concentrated load

The edge load developed a strong diagonal buttress and side-oriented reinforcement, achieving high robustness in the comparison.

Design insight

The loading study demonstrates that topology optimization is highly sensitive to force placement. A single point load concentrates material around the shortest force path, while distributed and edge loads encourage wider, more redundant structural systems. For design use, this means load assumptions must be defined early and tested carefully before interpreting the final form as a product-ready geometry.

OUTCOME & REFLECTION

Key outcomes

- Elongated aspect ratios produced the clearest stiffness-to-weight advantage because material could align directly with the main load path.
- Balanced geometries offered more stable and redundant load transfer but did not outperform the elongated domain in compliance behaviour.
- Tall/slim geometries were more bending-sensitive and required additional bracing, reducing overall efficiency.
- Distributed and edge loads produced broader, more robust support structures than a single central point load.
- Filtering and careful mesh setup were necessary to reduce checkerboard effects, mesh dependency, and unrealistic density artifacts.

Design value

This project demonstrates the overlap between engineering simulation and design decision-making. Instead of presenting topology optimization as a purely technical calculation, the case study communicates how computational outputs can guide form development, lightweighting, structural reasoning, and future additive manufacturing workflows.

What I would improve next

- Convert voxel-based outputs into smoother CAD-ready surfaces for presentation and manufacturability review.
- Add stress and displacement plots beside each topology to strengthen engineering validation.
- Run manufacturability-aware constraints for additive manufacturing, including minimum feature size and support reduction.
- Create a final product application, such as a bracket, support frame, or lightweight structural module, using the optimized design language.

Website-ready copy

Topology Optimization for 3D Structures Design is a computational design project that investigates how geometry and loading conditions influence lightweight structural forms. Using MATLAB, finite element analysis, and the SIMP method, I tested multiple 3D domains and loading profiles to understand how material distribution changes under different design constraints. The project demonstrates how simulation-led workflows can generate efficient structural concepts, reveal load paths, and support future design for additive manufacturing.

Short portfolio description

A 3D topology optimization study using MATLAB and SIMP to explore lightweight structural design under varying aspect ratios and load conditions.

Suggested project tags

Computational Design / Mechanical Engineering / Topology Optimization / MATLAB / FEA / Generative Design / Additive Manufacturing / Structural Design